Capstone Dataset and Baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One capstone design-and-baseline evidence record.

Define fields,audit source groups and preserve actual validation outputs.

Keep one habit: W1–W3/W5 → selected pair-plan task → exact project data bundle → Week35comparison.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Model Cards and Human Review

> **My remembered idea, example and correction**

Subgroup Performance

> **My remembered idea, example and correction**

Learning Curves

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Actual72-record kit:40 training parcels(22 clear/18 blocked), 20 validation(9 clear/11 blocked), 12 finalfeature-only rows; singletonsourcegroupIDs P00–P71. Train-majority clear givesvalidation 9/20=45%, blockedfound 0/11; positiveprecision unavailable. Student full CSV/program has blank finalreferences. IDs/groups/splits are metadata, width/heightcm candidatefeatures.

  Separate illustrative 30 clear/10 blockedtraining and 14/6 validation givesbaseline 70%, candidate 80%; those are not actual kit scores. Picture8 plants×3 views=24, developmentA–F18 views, finalG/H6 views, with training/validationinside development still to be specified.

- Write a bounded question and a non-use statement for the supplied task.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Actual72-record kit:40 training parcels(22 clear/18 blocked), 20 validation(9 clear/11 blocked), 12 finalfeature-only rows; singletonsourcegroupIDs P00–P71. Train-majority clear givesvalidation 9/20=45%, blockedfound 0/11; positiveprecision unavailable. Student full CSV/program has blank finalreferences. IDs/groups/splits are metadata, width/heightcm candidatefeatures.

  Separate illustrative 30 clear/10 blockedtraining and 14/6 validation givesbaseline 70%, candidate 80%; those are not actual kit scores. Picture8 plants×3 views=24, developmentA–F18 views, finalG/H6 views, with training/validationinside development still to be specified.

- Explain why record ID, group ID and measured width have different roles.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Actual72-record kit:40 training parcels(22 clear/18 blocked), 20 validation(9 clear/11 blocked), 12 finalfeature-only rows; singletonsourcegroupIDs P00–P71. Train-majority clear givesvalidation 9/20=45%, blockedfound 0/11; positiveprecision unavailable. Student full CSV/program has blank finalreferences. IDs/groups/splits are metadata, width/heightcm candidatefeatures.

  Separate illustrative 30 clear/10 blockedtraining and 14/6 validation givesbaseline 70%, candidate 80%; those are not actual kit scores. Picture8 plants×3 views=24, developmentA–F18 views, finalG/H6 views, with training/validationinside development still to be specified.

- Name three acceptance criteria that do not require the candidate to win.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual72-record kit:40 training parcels(22 clear/18 blocked), 20 validation(9 clear/11 blocked), 12 finalfeature-only rows; singletonsourcegroupIDs P00–P71. Train-majority clear givesvalidation 9/20=45%, blockedfound 0/11; positiveprecision unavailable. Student full CSV/program has blank finalreferences. IDs/groups/splits are metadata, width/heightcm candidatefeatures.

  Separate illustrative 30 clear/10 blockedtraining and 14/6 validation givesbaseline 70%, candidate 80%; those are not actual kit scores. Picture8 plants×3 views=24, developmentA–F18 views, finalG/H6 views, with training/validationinside development still to be specified.

- What should the team do if it cannot obtain a valid untouched final test?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- The selected 80-row/40-related-pair task differs from the kit's72 single-record parcels. In this task both rows in each pair share one source group, so assign whole pairs to roles and exclude pair ID from predictive inputs. An exact 24/8/8 pair allocation is one possible declared example:48/16/16 rows, provided coverage and intended-use requirements are checked.

  No class counts or baseline score are supplied; do not invent them.

- A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Define fields for Capstone data and **requirement**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.
- Plan grouping and **baseline**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Your capstone dataset includes duplicate photographs in both **training** and validation. What should be repaired before measuring the **baseline**?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 34 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

My reasoning and evidence

▼ **Extension · Capstone models and baselines — W1 Build the baseline**

REFERENCE · SECTION 17

Extension · Capstone models and baselines — W1 Build the baseline

> Retained extension task with its own values and teacher key.

Extension question and evidence

- State the worked **training**-majority rule and validation accuracy.

> **My reasoning and evidence**

▼ **Extension · Capstone models and baselines — W2 Evaluate the candidate**

REFERENCE · SECTION 18

Extension · Capstone models and baselines — W2 Evaluate the candidate

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Compute the worked candidate accuracy, precision and recall.

My reasoning and evidence

▼ Extension · Capstone models and baselines — W3 Explain the tradeoff

REFERENCE · SECTION 19

Extension · Capstone models and baselines — W3 Explain the tradeoff

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Compare errors rather than only headline accuracy.

My reasoning and evidence

▼ Extension · Capstone models and baselines — W4 Apply independently

REFERENCE · SECTION 20

Extension · Capstone models and baselines — W4 Apply independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- **Training** 18 clear/6 blocked; validation **baseline**[[9, 0],[3, 0]], candidate[[8, 1],[1, 2]]. Calculate accuracy for each and candidate precision/recall.

My reasoning and evidence

▼ **Extension · Capstone models and baselines — W5 Freeze evidence**

REFERENCE · SECTION 21

Extension · Capstone models and baselines — W5 Freeze evidence

> Retained extension task with its own values and teacher key.

Extension question and evidence

- List what must be saved before the teacher releases final-test labels.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Plan the plant capstone around independent groups and a **baseline** before collecting more rows.
- Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current **milestone**: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Plan the plant capstone around independent groups and a **baseline** before collecting more rows.
- Which labelled step matches this detail? Fit majority label only from development **training**; record metrics and error types.
- 1. Define target
- 2. Split by groups

- 3. Plan **baseline**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Plan **baseline** Fit majority label only from development **training**; record metrics and error types. Read the picture in order.

Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A derived crop of plantG is added to **training**. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The final group separation is violated.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The final group separation is violated. The crop retains the same source plant identity even with a new file name.

Repair and keep the record

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Capstone Dataset and Baseline: the case

AI ACADEMY / AI-4/34 · 01 · LOOK AT THE FACTS

Capstone Dataset and Baseline: the case

The situation

Plan the plant capstone around independent groups and a baseline before collecting more rows.

Supplied evidence

Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain capstone dataset and baseline using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

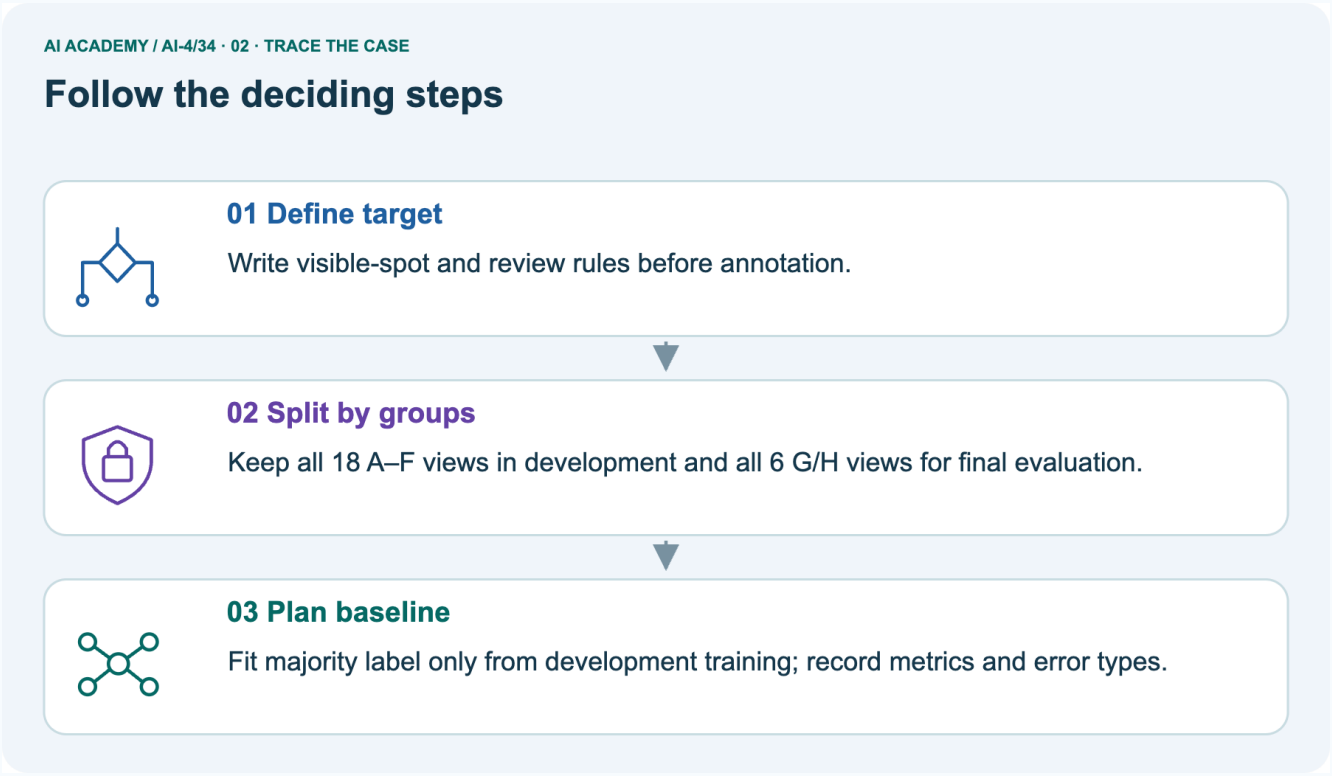
Plan the plant capstone around independent groups and a **baseline** before collecting more rows. Supplied facts: Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H.

Target is visible-spot status, with review for hidden leaves. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: Write visible-spot and review rules before annotation. → Split by groups: Keep all 18 A–F views in development and all 6 G/H views for final evaluation. → Plan **baseline**: Fit majority label only from development **training**; record metrics and error types.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Four-step worked evidence for Week 34: Capstone Dataset and

Baseline

AI ACADEMY / AI-4/34 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

Step	Evidence and reasoning
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 34: Capstone Dataset and **Baseline**. 1 | **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.. 2 | Test set remains sealed; validation supports model choices.. 3 | Validation has 12 healthy and 8 spotted leaves.

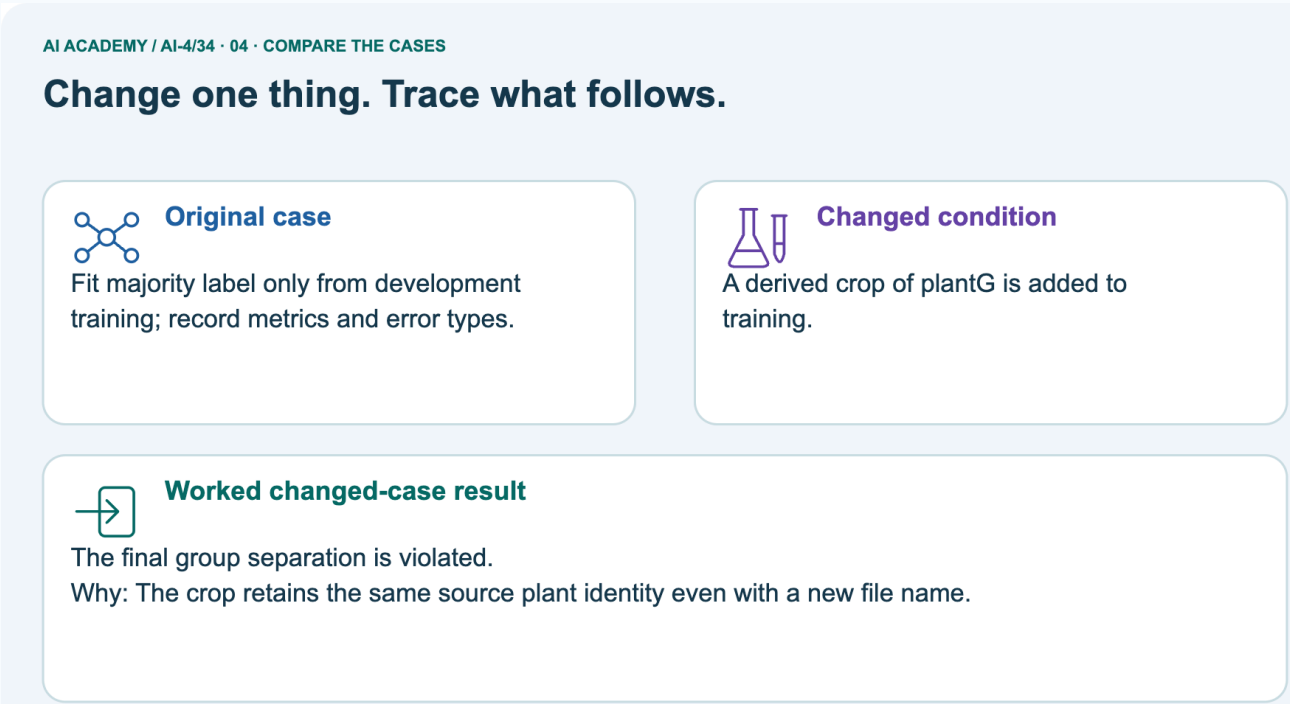
The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall.

The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.. 4 | Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

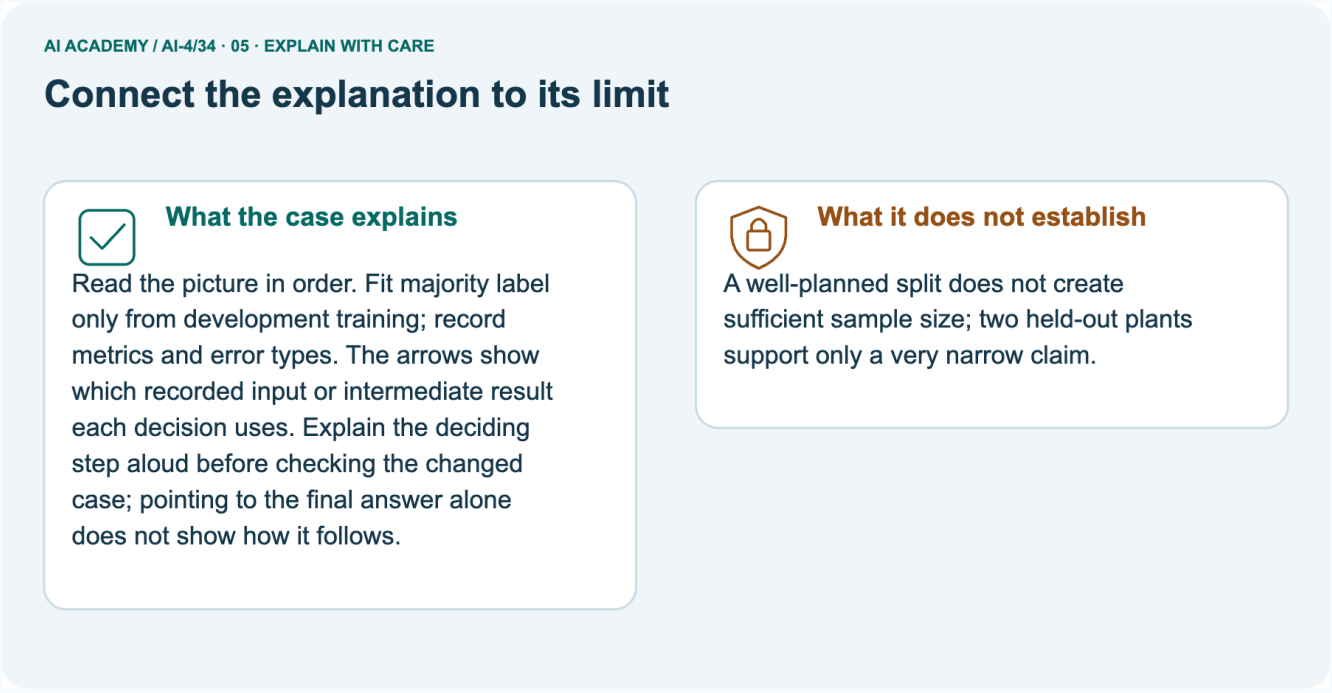
Original case: Fit majority label only from development **training**; record metrics and error types. Changed condition: A derived crop of plantG is added to **training**. Expected result: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ **Earlier wording, alternatives and picture transcripts**

01. Capstone Dataset and Baseline: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 34: Capstone Dataset and Baseline

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.